



Started on	Thursday, 17 September 2026, 10:38 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:39 AM
Time taken	1 min 13 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int binarySearch(int a[], int low, int high, int x, int index) {
4      if (low > high)
5          return -1;
6
7      int mid = (low + high) / 2;
8
9      if (a[mid] == x && mid != index)
10         return mid;
11
12     if (a[mid] < x)
13         return binarySearch(a, mid + 1, high, x, index);
14     else
15         return binarySearch(a, low, mid - 1, x, index);
16 }
17
18 int findPair(int a[], int n, int x, int i) {
19     if (i >= n)
20         return 0;
21
22     int target = x - a[i];
23
24     int j = binarySearch(a, 0, n - 1, target, i);
25
26     if (j != -1) {
27         printf("%d\n", a[i]);
28         printf("%d\n", a[j]);
29         return 1;
30     }
31
32     return findPair(a, n, x, i + 1);
33 }
34
35 int main() {
36     int n, x;
37     scanf("%d", &n);
38
39     int a[n];
40
41     for (int i = 0; i < n; i++)
42         scanf("%d", &a[i]);
43
44     scanf("%d", &x);
45
46     if (!findPair(a, n, x, 0))
47         printf("No");
48
49     return 0;
50 }

```

	Input	Expected	Got	
✓	4	4	4	✓
	2	10	10	
	4			
	8			
	10			
	14			
✓	5	No	No	✓
	2			
	4			
	6			
	8			
	10			
	100			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Jayashna A H 251801112-2025-AIDS ▾

J2

Started on	Thursday, 17 September 2026, 10:37 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:37 AM
Time taken	40 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int findFloor(int a[], int low, int high, int x) {
4     if (low > high)
5         return -1;
6
7     int mid = (low + high) / 2;
8
9     if (a[mid] == x)
10        return a[mid];
11
12    if (a[mid] > x)
13        return findFloor(a, low, mid - 1, x);
14
15    int result = findFloor(a, mid + 1, high, x);
16
17    if (result != -1)
18        return result;
19
20    return a[mid];
21 }
22
23 int main() {
24     int n, x;
25     scanf("%d", &n);
26
27     int a[n];
28
29     for (int i = 0; i < n; i++)
30         scanf("%d", &a[i]);
31
32     scanf("%d", &x);
33
34     printf("%d", findFloor(a, 0, n - 1, x));
35
36     return 0;
37 }
```

	Input	Expected	Got	
✓	6	2	2	✓
	1			
	2			
	8			
	10			
	12			
	19			
	5			
✓	5	85	85	✓
	10			
	22			
	85			
	108			
	129			
	100			
✓	7	9	9	✓
	3			
	5			
	7			
	9			
	11			
	13			
	15			
	10			

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 17 September 2026, 10:34 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:36 AM
Time taken	2 mins 19 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

Example 1:

Input: `nums = [3,2,3]`

Output: `3`

Example 2:

Input: `nums = [2,2,1,1,1,2,2]`

Output: `2`

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 104`
- `-231 <= nums[i] <= 231 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2
3 int majorityElement(int a[], int low, int high) {
4     if (low == high)
5         return a[low];
6
7     int mid = (low + high) / 2;
8
9     int left = majorityElement(a, low, mid);
10    int right = majorityElement(a, mid + 1, high);
11
12    if (left == right)
13        return left;
14
15    int countLeft = 0, countRight = 0;
16
17    for (int i = low; i <= high; i++) {
18        if (a[i] == left)
19            countLeft++;
20        if (a[i] == right)
21            countRight++;
22    }
23
24    return (countLeft > countRight) ? left : right;
25 }
26
```

```
27 | int main() {
28 |     int n;
29 |     scanf("%d", &n);
30 |
31 |     int a[n];
32 |
33 |     for (int i = 0; i < n; i++)
34 |         scanf("%d", &a[i]);
35 |
36 |     printf("%d", majorityElement(a, 0, n - 1));
37 |
38 |     return 0;
39 | }
```

	Input	Expected	Got	
✓	3 3 2 3	3	3	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 17 September 2026, 10:26 AM
State	Finished
Completed on	Thursday, 17 September 2026, 10:45 AM
Time taken	18 mins 28 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```

1 |
2 |
3 | #include <stdio.h>
4 |
5 | int main() {
6 |     int m, i, count = 0;
7 |     scanf("%d", &m);
8 |
9 |     int a[m];
10 |
11 |     for (i = 0; i < m; i++) {
12 |         scanf("%d", &a[i]);
13 |     }
14 |
15 |     for (i = 0; i < m; i++) {
16 |         if (a[i] == 0)
17 |             count++;
18 |     }
19 |
20 |     printf("%d", count);
21 |
22 |     return 0;
23 | }
```

	Input	Expected	Got	
✓	5 1 1 1 0 0	2	2	✓

	Input	Expected	Got	
✓	10 1 1 1 1 1 1 1 1 1 1 1 1	0	0	✓
✓	8 0 0 0 0 0 0 0 0 0 0	8	8	✓
✓	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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